

LUMINA AI

(ES-451: Minor Project)

submitted in partial fulfillment of the requirement
for the award of the degree of

**Bachelor of Technology
In
Computer Science Engineering**

Submitted by

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2022-2026



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DECLARATION

We, students of B. Tech hereby declare that the minor project titled “**Lumina AI**” which is submitted in department of Computer Science and Engineering, HMR Institute Of Technology and Management, Hamidpur, Delhi, affiliated to Guru Gobind Singh Indraprastha University, Dwarka, New Delhi in partial fulfillment of requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering has not been previously the basis for award of any degree, diploma or other similar title or recognition. The list of the members involved in the project is listed below:

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This is to certify that this project report entitled “**LUMINA AI**” submitted by **Ayush Kumar (35113302722)**, **Nandan Pradhan (08013302722)** and **Nishant Gupta (04113302722)** and submitted to Department of Computer Science and Engineering, HMR Institute of Technology & Management, Hamidpur Delhi, affiliated to Guru Gobind Singh Indraprastha University, Dwarka(New Delhi) in partial fulfillment of requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering, during the academic year 2025-26, is a Bonafide record of work carried out under our guidance and supervision from **August to December 2025**.

The results embodied in this report have not been submitted to any other University or Institution for the award of any degree or diploma.

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ACKNOWLEDGEMENT

The success and outcome of this project required a lot of guidance and assistance from many people and we are extremely privileged to have got this all along the completion of our project.

I wish to express my gratitude to **Prof. (Dr.) V. C Pandey Director, HMRITM Delhi** for providing the facilities of the Institute and for his encouragement during the course of this work. I also express our deep indebtedness to our mentor, **Ms. Usha Dhankar (Coordinator CSE DS)** for her guidance and constant supervision as well as for providing necessary information regarding the project in addition to offering her consistent support in completing the project.

I would also like to take this opportunity to acknowledge the guidance from **Dr. Naveen Sharma (Coordinator, CSE)** for his kind cooperation and encouragement which helped us in the successful completion of this project.

In addition to the aforementioned, I would like to thank Project Coordinator **Dr. Naveen Sharma (Coordinator, CSE)** for providing us with guidance and help on every step of the way during the project and for imparting us with invaluable knowledge and teaching us the etiquette of a research student. Finally, I wish to thank my family members and my friends who have always been very supportive and encouraging.

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ABSTRACT

Prolonged exposure to digital screens has led to a sharp rise in Digital Eye Strain or Computer Vision Syndrome (CVS), especially among students, professionals and remote workers. Existing wellness tools rely on rigid timers or generic alerts that are often intrusive and quickly dismissed, provide limited medical value. This report represents project Lumina AI, a cross- platform desktop application designed to deliver intelligent, real-time eye-health monitoring and personalized wellness interventions. Built with an Electron-based frontend and a Python powered computer vision backend, Lumina AI uses OpenCV and dlib facial landmark detection to compute the Eye Aspect Ratio (EAR) for precise blink-frequency tracking with over 85% detection accuracy and sub-50 ms latency. A multi-model reminder engine issues visual, auditory and contextual prompts, while a personalized module adapts intervals, display options and interaction styles to individual user behavior. Advanced features such as 20-20-20 exercise guidance and Meibomian Gland Dysfunction (MGD) support extend its clinical relevance. Experimental evaluation shows Lumina AI achieves lightweight performance (<100 MB RAM, <5% CPU) across varied lighting conditions, offering a scalable and privacy-friendly foundation for future integration with wearables, AR/VR platforms and workplace health programs.

Keywords: Computer Vision, Eye Detection, Blink Monitoring, Digital Wellness, Machine Learning, RealTime Systems, Human-Computer Interaction

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CHAPTER 1

INTRODUCTION

The rapid digitization of modern workplaces has fundamentally transformed professional technology interaction patterns, with knowledge workers averaging 7-9 hours daily before digital displays. This unprecedented screen exposure has given rise to Computer Vision Syndrome (CVS), a constellation of visual and ergonomic symptoms affecting over 60% of computer users globally. Clinical manifestations include dry eyes, blurred vision, headaches, neck pain, and general ocular discomfort that significantly impacts productivity and quality of life.

Research indicates that blink rates decrease dramatically from the normal 15-20 blinks per minute to as low as 5 blinks per minute during concentrated computer use. This physiological change causes inadequate tear film distribution across the corneal surface, leading to evaporative dry eye and associated discomfort. The tear film's three-layer structure—comprising an outer lipid layer preventing evaporation, a middle aqueous layer providing nutrients and oxygen, and an inner mucin layer ensuring proper distribution—becomes disrupted when blinking frequency decreases, accelerating tear evaporation and corneal exposure.

The American Optometric Association estimates that 83% of Americans use digital devices for more than two hours daily, with 53.1% experiencing eye strain symptoms and 27.2% reporting persistent symptoms even after device cessation. The COVID-19 pandemic accelerated remote work adoption, increasing average screen time from 6.5 hours in 2019 to over 9 hours in 2024, correlating with a 38% rise in reported eye strain symptoms across multiple demographics.

Lumina AI: Intelligent Eye Health Monitoring

Lumina AI addresses this growing healthcare challenge through innovative computer vision technology that monitors blink patterns in real-time and provides intelligent, adaptive interventions. Unlike traditional timer-based reminder applications that operate on rigid schedules regardless of actual user physiology, Lumina AI shifts to adaptive health monitoring that responds directly to physiological indicators detected through advanced facial analysis.

The system employs 68-point facial landmark detection using the dlib machine learning library, combined with Eye Aspect Ratio (EAR) calculations for precise blink detection across diverse environmental conditions. This mathematical approach, validated in computer vision research by Soukupová & Čech (2016), provides robust detection accuracy even with variations in lighting, head pose, and individual facial anatomy.

Built on the Electron framework, Lumina AI delivers cross-platform desktop performance seamlessly across Windows and macOS operating systems from a unified codebase. This architectural decision ensures consistent user experience while simplifying maintenance and feature development. The system architecture prioritizes privacy through complete local processing—all computer vision operations execute on-device with no internet connectivity required, ensuring that no personal health data or biometric information ever leaves the user's device.

The system integrates evidence-based medical protocols derived from peer-reviewed ophthalmology research. Support for Meibomian Gland Dysfunction (MGD)—a condition affecting up to 86% of certain demographics—provides automated therapeutic reminders following clinical protocols. The clinically-validated 20-20-20 rule (every 20 minutes, look at something 20 feet away for 20 seconds) is implemented with automated compliance tracking, bridging the gap between clinical research recommendations and practical consumer wellness technology.